Animesh Joshi

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Education

Purdue University, West Lafayette West Lafayette, Indiana
B.S. in Mathematics and Statistics *Expected Dec 2025*
Courses: Multivariate Stats, Design of Experiments, Stochastic Processes, Advanced Linear Algebra, Statistical Theory, Operations Research, Probability, Regression Analysis, Machine Learning, Causal Inference (Coursera)

Experience

Software Systems Engineer Intern May 2025 - Aug 2025
IT/OT Department; Dexcom *San Diego, Ca*

- Improved the precision of performance verification testing (PVT) for 10k+ continuous glucose monitors by rigorously testing data analysis software used to process raw sensor data and produce estimates of sensor accuracy
- Built Python-based data generation tools to simulate raw sensor datasets with customizable statistical properties (error distributions, tolerance/Chebyshev bounds, noise, signal spikes), enabling detection of software defects and statistical inaccuracies in performance verification testing software
- Validated statistical computations in performance verification software by developing python scripts to process generated raw sensor waveforms and compute results for normality tests (Shapiro–Wilk, Anderson–Darling, Skewness–Kurtosis), tolerance intervals (two-sided, distribution-free), and Chebyshev bounds
- Updated automated testing application for SQL stored procedures used in manufacturing 1M+ continuous glucose monitors by developing Python libraries and SQL queries to enhance code maintainability.
- Upgraded Python-based report generation application for automated software tests, enhancing readability, adding new report fields, and enabling report generation for 100k+ test cases

Credit Analytics Intern Jul 2024 - Aug 2024
Credit Analytics Department; Axos Bank *San Diego, Ca*

- Developed SQL scripts to extract and aggregate loan-level data for 37,000+ loans from the Enterprise Data Warehouse, creating datasets with key risk metrics including LTV, FICO Scores, and Delinquent Balances
- Used SQL to analyze loan-level datasets and generate summary tables tracking monthly trends in the quality of 4,000+ new commercial loan originations over the past fiscal year and overall asset quality in the bank's loan portfolio, supporting credit risk monitoring and strategic decision-making
- Built a Tableau workbook visualizing credit risk trends over time for $12.5 billion in recent loan originations and $20 billion in active loans, supporting investigation into factors contributing to $500 million in delinquent balances

Data Science Researcher Aug 2022 - May 2023
The Data Mine; Purdue University *West Lafayette, IN*

- Enabled automatic detection of algal contamination in waterbodies by fine-tuning a ResNet50 image classification model on satellite imagery data, achieving 86% accuracy in identifying contaminated sites
- Developed a U-Net algorithm for semantic segmentation of satellite imagery to automatically generate algal contamination masks for waterbodies, achieved 70% similarity between ground-truth masks and syntethic masks

Selected Projects

Recidivism Risk Forecasting
National Institute of Justice

- Leveraged statistical learning algorithms (regression, discriminant analysis, random forests, gradient boosting) to forecast recidivism probabilities for prisoners in Georgia using data compiled by the National Insitute of Justice
- Used K-Fold cross validation algorithm to generate sampling distributions of various prediction metrics and used generated distributions to test machine learning algorithms for unfair treatment of African-American prisoners
- Evaluated fairness definitions based on statistical disparities in model predictions for protected and privileged groups using t-tests. Assessed causal fairness by applying McNemar's test to contingency tables generated from counterfactual and nearest neighbor matching

Professional Techniques

Programming Languages: Python, SQL, R, SAS, Matlab
Packages/Tools: Jupyter, R-Studio, VSCode, Git, SSMS, Pandas, Sci-Kit Learn, Tensorflow, Tableau
Skills: Machine Learning, Statistics, Causal Inference, Software Engineering, Experimentation, Database Management